

For Sequential cells, the following characterizations have to be performed and filled.

a) Input pin capacitances:

Input Pins	Rise Cap (fF)	Fall Cap (fF)	Average Cap (fF)
D	4.01855	4.01825	4.0184
CLK	20.9715	20.9719	20.9717

b) Set-up Time Table:

(i) Rise Constraint (in ns) [Input slew vs CLK slew].

	10 ps	1000 ps
10 ps	0.152	0.333
1000 ps	0.243	0.409

(ii) Fall Constraint (in ns) [Input slew vs CLK slew].

	10 ps	1000 ps
10 ps	0.158	0.355
1000 ps	0	0.17

c) Hold Time Table:

(i) Rise Constraint (in ns) [Input slew vs CLK slew].

	10 ps	1000 ps
10 ps	0	0
1000 ps	0.034	0

(ii) Fall Constraint (in ns) [Input slew vs CLK slew].

	10 ps	1000 ps
10 ps	0	0
1000 ps	0	0

d) Transition Time Table (for Q output, CLK will have minimum slew of 10 ps): (please strictly consider 20% and 80% of VDD for transition time)

(i) Output Rise Transitions (in ns) [Input slew vs output capacitance].

Related pin D: (i.e., other input pins are held constant)

	10 ps	100 ps	1000 ps
0.5 fF	0.018	0.018	0.018
10 fF	0.068	0.068	0.068
100 fF	0.596	0.596	0.596

(ii) Output Fall Transitions (in ns) [Input slew vs output capacitance].

Related pin D: (i.e., other input pins are held constant)

	10 ps	100 ps	1000 ps
0.5 fF	0.027	0.027	0.027
10 fF	0.083	0.083	0.083
100 fF	0.644	0.644	0.644

e) **CLK-to-Q Delay Time Table:** (delay between clock transition and data transition. Use 50% of CLK to 50% of output to simulate propagation delay).

(i) Cell Rise Delay (in ns) [Input slew vs output capacitance].

Related pin D: (i.e., other input pins are held constant)

	10 ps	100 ps	1000 ps
0.5 fF	0.196	0.196	0.196
10 fF	0.244	0.244	0.244
100 fF	0.596	0.596	0.596

(ii) Cell Fall Delay (in ns) [Input slew vs output capacitance].

Related pin D: (i.e., other input pins are held constant)

	10 ps	100 ps	1000 ps
0.5 fF	0.288	0.288	0.288
10 fF	0.346	0.346	0.346
100 fF	0.785	0.785	0.785

f) **Static Power** (all possible input combinations of CLK and D).

Condition (CLK, D)	Power (nW)
00	4.226
01	5.252
10	0.429
11	0.612

g) **Dynamic Power Table:** (CLK will have minimum slew of 10 ps)

(i) Rise Power (in nW) [Input slew vs output capacitance].

Related pin D: (i.e., other input pins are held constant)

	10 ps	100 ps	1000 ps
0.5 fF	37195.4	37038.6	37098.6

10 fF	44971.4	44814.6	44870.9
100 fF	117642	117485	117541

(ii) **Fall Power (in nW)** [Input slew vs output capacitance].

Related pin D: (i.e., other input pins are held constant)

	10 ps	100 ps	1000 ps
0.5 fF	42921.7	42821.2	42862.8
10 fF	43065.5	42965.1	43007.3
100 fF	43197.4	43097	43137.7